

Problem 5.1 Copper plate is bent into a circular arc of radius 10 meters. If the plate section be 120 mm wide and 20 mm thick, find the maximum stress induced and the bending moment which can produce this stress. Take $E = 1 \times 10^5 \text{ N/mm}^2$.

Solution: Moment of inertia of the section about the neutral axis.

$$I = \frac{12}{120 \times 20} = 80000 \text{ mm}^4$$

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \cdot y$$

$$\therefore \sigma_{\max} = \frac{1 \times 10^5}{10 \times 10^3} \times \left(\frac{20}{2} \right) \text{ N/mm}^2 = 100 \text{ N/mm}^2$$

$$M = \frac{E}{R} \cdot I = \frac{1 \times 10^5}{10 \times 10^3} \times 8 \times 10^4 = 8 \times 10^5 \text{ N mm}$$

$$= 800 \text{ Nm}$$

Problem 5.2 A timber joist of 200 mm width and 400 mm depth is used as a cantilever beam of 3 meters length and loaded with a concentrated load at its free end in addition to its self-weight, so that the bending stress does not exceed 7 N/mm^2 . Find out the maximum value of the applied load. Unit weight of timber is 5 kN/m^3 .

Solution: Let W (N) be the concentrated load that can be placed safely at the free end of the cantilever under the given conditions of the problem. W = self weight of the joint per meter run.

$$= 0.2 \times 0.4 \times 1 \times 5 = 0.4 \text{ kN/m} = 400 \text{ N/m}$$

I_{xx} = Moment of inertia of the section about N.A.

$$= \frac{12 \times 200 \times 400^3}{32} = 32 \times 10^8 \text{ mm}^4$$

M = Maximum bending moment which will occur at the fixed end.

$$= WL + \frac{WL^2}{2}$$

$$= W \times 3 + \frac{100 \times 3 \times 3}{2} \text{ Nm} = (3W + 450) \times \frac{10^3}{400} \text{ N mm}$$

$$y = \text{Distance of extreme fiber from N.A.} = \frac{400}{2} = 200 \text{ mm.}$$

$$\sigma = \text{Bending stress} = 7 \text{ N/mm}^2$$

Using the relation $\frac{M}{I} = \frac{\sigma}{y}$ and substituting the values

$$\frac{(3W + 450) \times 10^3 \times 3}{32 \times 10^8} = \frac{7}{200}$$

or

$$3W + 450 = 37333.33$$

$\therefore$

$$W = 12294.44 \text{ N}$$

Problem 5.3 A cast iron T-beam has the following dimensions:

Overall depth	= 160 mm
Width of flange	= 150 mm
Flange thickness	= 40 mm
Web thickness	= 50 mm

The beam is simply supported over a span of 2.5 meters placed in the inverted T-position (i.e. flange at the bottom). If the maximum allowable tensile stress in the flange and compressive stress in the web are limited to 25 N/mm² and 80 N/mm² respectively, find the maximum central load that the beam can safely carry.

Solution: As the Fig. 5.8 is symmetrical about Y-Y axis, the centre of gravity will lie on this axis. Let $\bar{y}$ be the distance of centre of gravity from the bottom of flange. Divide the figure into two rectangles as shown in Fig. 5.8.

Rectangle 1

$$a_1 = 150 \times 40 = 6000 \text{ mm}^2, y_1 = \frac{40}{2} = 20$$

mm

Rectangle 2

$$a_2 = 120 \times 50 = 6000 \text{ mm}^2, y_2 = 40 + \frac{120}{2}$$

= 100 mm

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{6000 \times 20 + 6000 \times 100}{6000 + 6000}$$

= 60 mm

Moment of inertia of the section about NA

$$= \frac{1}{12} \times 150 \times 40^3 + 150 \times 40 \times (60 - 20)^2 + \frac{1}{12} \times 50 \times 120^3 + 50 \times 120 \times (100 - 60)^2$$

$$= 272 \times 10^5 \text{ mm}^4$$

σ_t = Permissible tensile stress at the bottom most face of flange = 25 N/mm²,
at a distance of $y_t = 60$ mm from neutral axis.

σ_c = Permissible compressive stress at the top most face of web = 80 N/mm²,
at a distance of $y_c = 100$ mm from neutral axis.

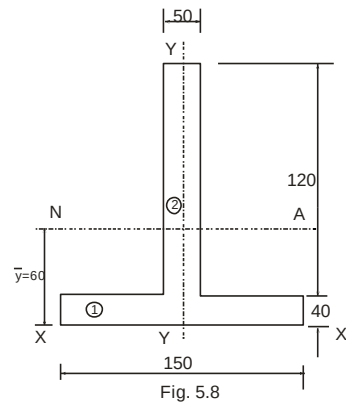
$$\frac{\sigma_t}{y_t} = \frac{25}{60} = 0.42, \quad \frac{\sigma_c}{y_c} = \frac{80}{100} = 0.80$$

$\frac{\sigma_t}{y_t}$

y_t being the least, is to be taken into consideration. Let W N be the concentrated load

that can be safely carried under the given condition of the problem. Maximum bending

moment at the centre, $M = \frac{Wl}{4} = \frac{4 \times 2.5}{4} \text{ Nm}$.



$$\frac{M}{I} = \frac{\sigma_t}{y_t} \text{ and substituting the values}$$

$$\frac{W \times 2.5 \times 1000}{4 \times 272 \times 10^5} = 0.42$$

$$\therefore w = 18278.4 \text{ N}$$

Problem 5.4 A 4 meters long beam with rectangular section of 100 mm width and 200 mm depth is simply supported at the ends. If it is loaded with a uniformly distributed load of 4 kN/m throughout the span and a concentrated load $P = 2 \text{ kN}$ placed at a distance of 1.5 m from one end, determine the maximum bending stress in the beam.

Solution: Let R_A and R_B be the reactions at A and B respectively. Taking moment of all the forces about A:

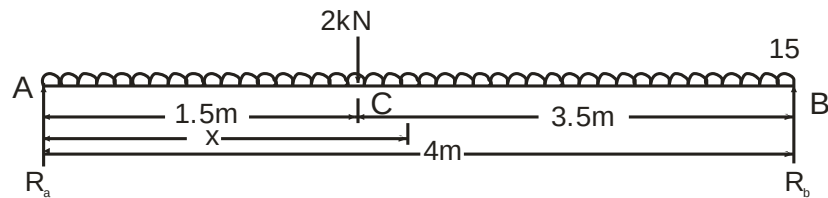


Fig. 5.9

$$R_B \times 4 - 4 \times 4 \times \frac{4}{2} - 2 \times 1.5 = 0 \quad \therefore R_B = 8.75 \text{ kN.}$$

$$\text{But } R_A + R_B = 4 \times 4 + 2 = 18$$

$$\therefore R_A = 18 - 8.75 = 9.25 \text{ kN.}$$

Let us consider a section at a distance x from A.

$$M_x = 9.25x - 2(x - 1.5) - 4 \cdot x \cdot \frac{x}{2} = 7.25x + 3 - 2x^2$$

M_x is maximum when

$$\frac{d}{dx} (7.25x + 3 - 2x^2) = 0$$

$$\text{or } 7.25 - 4x = 0 \quad \therefore x = 1.81 \text{ m}$$

$$\text{Moment bending of the section, } M = \frac{7.25 \times 1.81 + 3 - 2 \times (1.81)^2}{100 \times (200)} = 9.5703 \text{ kN.m}$$

$$\text{Moment of inertia of the section, } I = \frac{12}{100 \times (200)^3} = 6666.666 \times 10^4 \text{ mm}^4$$

$$\therefore \text{Maximum bending stress, } \sigma = \frac{M}{I} \times y = \frac{9.5703 \times 10^3 \times 10^3}{6666.66 \times 10^4} \times 100$$

$$= 14.35 \text{ N/mm}^2$$

Problem 5.5 A floor has to carry a load of 10000 N/sq.m. It is supported by joists 150 mm $\times$ 300 mm over a span of 5 m. How far apart may the joists be placed so that the bending stress does not exceed 8 N/mm²?

Solution: Let x be the centre-to-centre spacing of the joists. The area of the floor between any two joists = $5 \times x \text{ m}^2$.

The total load supported by one interior joist = $5 \times x \times 10000 = 50000x \text{ N}$.

$$\text{Maximum bending moment, } M = \frac{wl^2}{8} = \frac{50000x \times 5^2}{8} \text{ Nm.}$$

$$\text{Moment of inertia of the joist, } I = \frac{bd^3}{12} = \frac{0.15 \times (0.30)^3}{12} = 3.375 \times 10^{-4} \text{ m}^4$$

$$\frac{I}{y} \times \sigma = \frac{M}{y}$$

$$\text{We know, } M = \frac{50000x \times 5^2}{8} = 18000 \text{ Nm}$$

Equating this to the maximum bending moment

$$\frac{50000x \times 5^2}{8} = 18000$$

$$\therefore x = 0.576 \text{ m}$$

Problem 5.6 Compare the weight of two equally strong beams of circular sections made of the same material, one being of solid section and the other of hollow section with inside diameter $\frac{2}{5}$ of the external diameter.

Solution: For solid let the diameter be d

For hollow section: Let d_1 = Outside diameter

d_2 = Inside diameter

$$\therefore \frac{d_1}{d_2} = \frac{5}{2}$$

$$\text{For solid section, } Z_s = \frac{\pi d^3}{32}$$

$$\frac{\pi(d_1^4 - d_2^4)}{32d_1} = \frac{\pi d_1^3}{32} \left[1 - \left(\frac{2}{5} \right)^4 \right] = \frac{609\pi d_1^3}{32 \times 625}$$

$$\text{For hollow section, } Z_H =$$

$$\text{As both the beams are equally strong, } Z_s = Z_H$$

$$\frac{\pi d^3}{32} = \frac{609\pi d_1^3}{32 \times 625} \quad \text{or} \quad \frac{d_1^3}{d^3} = \frac{625}{609}$$

$$\therefore \frac{d_1}{d} = 1.00868$$

Both the beams are made of same material. Therefore $\rho_s = \rho_H = \rho$

$$\text{Let } W_s = \text{weight of solid shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} d^2 \times L \times \rho$$

$$W_H = \text{Weight of hollow shaft} = \text{volume} \times \text{density}$$

$$= \frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho$$

$$\therefore \frac{W_s}{W_H} = \frac{\frac{\pi}{4} d^2 \times L \times \rho}{\frac{\pi}{4} \left[d_1^2 - \left(\frac{2}{5} d_1 \right)^2 \right] \times L \times \rho} = \frac{d^2}{d_1^2 - \frac{4}{25} d_1^2}$$

$$\frac{W_s}{W_H} = \frac{25d^2}{21d_1^2} = \frac{25}{21} \times \frac{1}{(1.00868)^2} = 1.17$$

Therefore the weight of solid shaft is 1.17 times the weight of the hollow shaft.

Problem: 5.7 Fig. 5.10 shows an I-section proposed to be used for a beam of span 8 meters. The beam carries a uniformly distributed load of 100 kN/m over the whole span. Find the extreme bending stress for the section and state if the stress is permissible. If not, find the width of a 15 mm thick cover plate for each flange for the section to be safe. Find also the length of this cover plate. Safe stress in bending is 160 N/mm².

Solution: Moment of inertia of the I-section³

$$= I = \frac{250 \times 640^3}{12} - \frac{235 \times 600^3}{12} \text{ mm}^4$$

$$= 1.2313 \times 10^9 \text{ mm}^4$$

Maximum bending moment for the beam

$$= M = \frac{wl^2}{8} = \frac{100 \times 8^2}{8} = 800 \text{ kNm}$$

Extreme stress due to bending moment

$$= f = \frac{M}{I} y = \frac{800 \times 10^6}{1.2313 \times 10^9} \times 320 \text{ N/mm}^2$$

$$= 208 \text{ N/mm}^2$$

This is greater than the permissible stress of 160 N/mm². Suppose a cover plate of 15 mm thickness of provided for each flange, Increase in moment of inertia

$$= 2 \left[\frac{b \times 15^3}{12} + b \times 15 \times 327.5^2 \right]$$

$$= 3218250 b \text{ mm}^4$$

Moment of inertia of the modified section

$$= I' = 1.2313 \times 10^9 + 3218250 b \text{ mm}^4$$

Limiting the extreme stress to the permissible value of 160 N/mm²

$$f = 160 = \frac{M}{I'} y'$$

$$\frac{800 \times 10^6 \times 327.5}{I'}$$

$$\therefore I' = \frac{160}{160} = 1.63752 \times 10^9 \text{ mm}^4$$

$$\therefore 1.2313 \times 10^9 + 3218250 b = 1.63752 \times 10^9$$

$$\therefore b = 126.2 \text{ mm say } 130 \text{ mm}$$

Moment of resistance of the I section alone

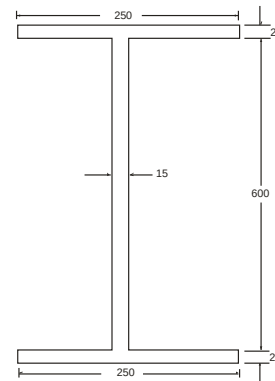


Fig. 5.10

$$= \frac{I}{y} \times f = \frac{1.2313 \times 10^9}{320} \times 160 = 6.1565 \times 10^8 \text{ Nmm} = 615.65 \text{ kNm}$$

Let the bending moment for the girder reach 615.65 kNm at a distance of x meter from the support

$$\therefore \frac{100 \times 10}{2} x - 100 \times \frac{x^2}{2} = 615.65$$

$$\therefore 50x^2 - 500x + 615.65 = 0$$

$$\therefore x^2 - 10x + 12.31 = 0 \quad \therefore 1.44 \text{ m from each end.}$$

Hence length of cover plate required

$$\therefore = 10 - 1 \times 1.44 = 7.124 \text{ m}$$

Problem 5.8 A timber beam is freely supported on supports 6 meters apart. It carried a uniformly distributed load of 12 kN per meter run and a concentrated load of 9 kN at 2.5 meters from the left support. If the stress in timber is not to exceed 10 N/mm², design a suitable section making the depth twice the width.

Solution: Fig. 5.11 shows the beam carrying the loading mentioned in the problem. Let R_a and R_b be the reactions at the left and right supports.

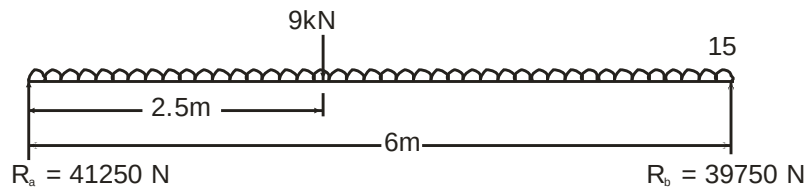


Fig. 5.11

Taking moments about the left support A, we have

$$R_b \times 6 = 12000 \times 6 \times 3 + 9000 \times 2.5$$

$$\therefore R_b = 39750 \text{ N}$$

$$\therefore R_a = 12000 \times 6 + 9000 - 39750 \text{ N} = 41250 \text{ N}$$

Let the shear force be zero at x meter from B

$$\therefore 39750 - 12000x = 0$$

$$\therefore x = \frac{39750}{12000} = 3.3125 \text{ m}$$

$\therefore$ Maximum bending moment occurs at 3.3125 meters from B

$$\therefore \text{Maximum bending moment} = 39750 \times 3.3125 - \frac{12000 \times (3.3125)^2}{2} \text{ Nm}$$

$$= 65836 \text{ Nm} = 65836 \times 1000 \text{ Nmm}$$

But the moment of resistance of the section

$$= \frac{1}{6} \sigma_b b d^2$$

Since $\sigma_b = 10 \text{ N/mm}^2$ and $b = \frac{d}{2}$, we have,

$$\frac{1}{6} \times 10 \times \frac{d}{2} \times d^2 = 65836 \times 1000$$

$$d^3 = \frac{65836 \times 1000 \times 12}{10}$$

$$\begin{aligned} \therefore d &= 369.64 \text{ mm} \cong 370 \text{ mm} \\ \therefore b &= \frac{d}{2} = \frac{370}{2} = 184.8 \text{ mm} \cong 185 \text{ mm} \\ \therefore \text{The beam section is } 231 \text{ mm} \times 462 \text{ mm} \end{aligned}$$

Problem 5.9 Find the width x of the flange of a cast iron beam having the section shown in Fig. 5.12 so that the maximum compressive stress is three times the maximum tensile stress the member being in pure bending subjected to sagging moment. The depth h of the beam is 80 mm, the thickness t of the web and flange is 20 mm.

Solution: Let the distance of the centroidal neutral axis from the bottom edge be $\bar{y}$. Let σ_c and σ_t be the maximum compressive and maximum tensile stress

$$\therefore \frac{\sigma_c}{\sigma_t} = \frac{\bar{y}}{h - \bar{y}} = 3$$

$$\therefore \frac{h - \bar{y}}{\bar{y}} = 3$$

$$\therefore \bar{y} = \frac{h}{4}$$

But from the geometry of the section, $\bar{y} = \frac{th \cdot \frac{h}{2} + (x-t)t \cdot \frac{t}{2}}{th + (x-t)t} = \frac{h}{4}$

$$\therefore x = t + \frac{h^2}{h - 2t}$$

Putting, $t = 20 \text{ mm}$, and $h = 80 \text{ mm}$

$$\therefore x = 20 + \frac{80^2}{80 - 2 \times 20} \text{ mm}$$

$$\therefore x = 180 \text{ mm}$$

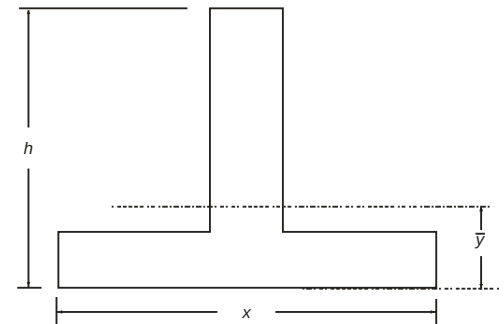


Fig. 5.12

Problem 5.10 A cast iron bracket subject to bending has cross-section of I-form with unequal flanges. The depth of the section is 280 mm and the metal is 40 mm thick throughout. The top flange is 200 mm wide and the bottom flange is 100 mm wide. Find the position of the neutral axis and the moment of inertia of the section about the neutral axis and determine the maximum bending moment that should be imposed on this section if the tensile stress in the top flange is not to exceed 20 N/mm^2 . What is then the value of the maximum compressive stress in the bottom flange?

Solution: Fig. 5.13 shows the section of the bracket. The section may be conveniently divided into three rectangular components. The areas of the individual components, their centroidal distance from the top edge and their moments about the top edge and their moments of inertia about their centroidal axes are tabulated below:

Component	Area a mm^2	Centroidal distance y from top edge y (mm)	ay (mm) ³	ay^2 (mm) ⁴	I_{self} (mm) ⁴
Top Flange	8000	20	16×10^4	320×10^4	$\frac{200 \times 40^3}{12} = 106.67 \times 10^4$
Web	8000	140	112×10^4	15680×10^4	$\frac{40 \times 200^3}{12} = 2666.67 \times 10^4$
Bottom Flange	4000	260	104×10^4	27040×10^4	$\frac{100 \times 40^3}{12} = 53.33 \times 10^4$
TOTAL	20000		232×10^4	43040×10^4	2826.67×10^4

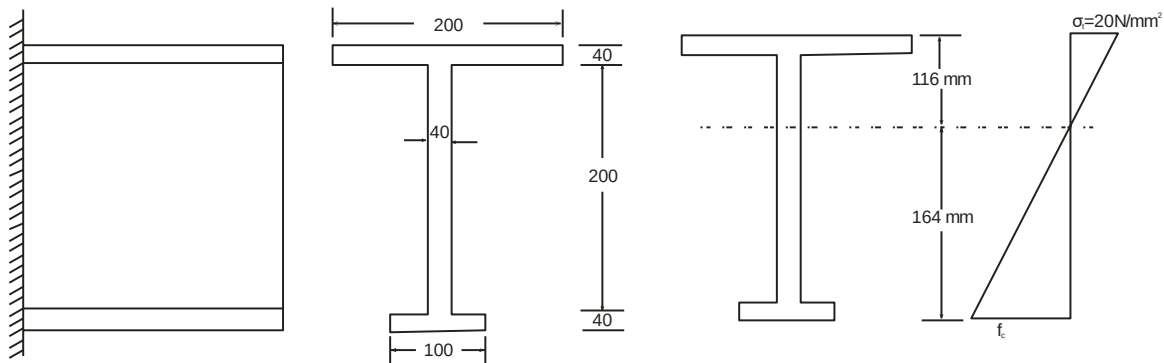


Fig. 5.13

$$\therefore \text{Distance of the neutral axis from the upper edge} \\ = \bar{y} = \frac{\sum ay}{\sum a} = \frac{232 \times 10^4}{20000} = 116 \text{ mm}$$

Moment of inertia about the upper edge

$$= I_{uu} = \sum I_{self} + \sum ay^2$$

$$2826.67 \times 10^4 + 43040 \times 10^4 = 45866.67 \times 10^4 \text{ mm}^4$$

But

$$I_{uu} = I_{xx} + (\sum a)\bar{y}^2$$

$$45866.67 \times 10^4 = I_{xx} + 20000 \times 116^2$$

$$\therefore I_{xx} = (45866.67 - 26912) \times 10^4 = 18954.67 \times 10^4$$

Let the maximum bending moment be in Nmm.

$$\frac{M}{I} = \frac{\sigma}{y_t}$$

$$\frac{\sigma_t}{y_t} \times I$$

$\therefore$

$$M = \frac{\sigma_t}{y_t} \times I \quad (\text{But } \sigma_t = 20 \text{ N/mm}^2)$$

$$= \frac{20}{116} \times 18954.67 \times 10^4 \text{ Nmm}$$

$$= 3268 \times 10^4 \text{ Nmm} = 32680 \text{ Nm}$$

Let the maximum compressive stress be $\sigma_c = \frac{164}{116} \times 20 \text{ N/mm}^2$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

$$\therefore \sigma_c = 28.28 \text{ N/mm}^2$$

Problem 5.11 The following three beams have the same length and are of the same material and have same weight.

- (i) I-section 400 mm × 180mm with 20 mm thick flanges and 12.5 mm thick web.
- (ii) Rectangular section of depth twice its width.
- (iii) Solid circular section.

Compare the strength of the I-section with the rectangular and circular sections.

Solution: Obviously all the three beams have the same sectional sections area.

Properties of the I-section

$$\text{Area} = 2(180 \times 20) + (360 \times 12.5) = 11700 \text{ mm}^2$$

$$= 3.0876 \times 10^8 \text{ mm}^3$$

$$\text{Section modulus } Z_1 = \frac{3.0876 \times 10^8}{200}$$

$$= 1543800 \text{ mm}^3$$

Properties of the rectangular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Width} = b, \text{ depth} = d = 2b$$

$$\therefore \text{Area} = 2b^2 = 11700$$

$$\therefore b = 76.485 \text{ mm}$$

$$\text{and } d = 2 \times 76.485 = 152.97 \text{ mm}$$

$$\text{section modulus } Z_2 = \frac{bd^3}{6} = \frac{76.485 \times 152.97^3}{6} = 298289.22 \text{ mm}^3$$

Properties of circular section

$$\text{Area} = 11700 \text{ mm}^2, \text{ Diameter} = d$$

$$\therefore \text{Area} = \frac{\pi d^2}{4} = 11700 \quad \therefore d = 122.053 \text{ mm}$$

$$\text{Section modulus } Z_3 = \frac{\pi d^3}{32} = \frac{\pi \times 122.053^3}{32} = 178502.9 \text{ mm}^3$$

$$\text{Comparison of strength} = \frac{\text{Strength of I-section}}{\text{Strength of rectangular section}} = \frac{Z_1}{Z_2} = \frac{1543800}{298289.22} = 5.176$$

$$\frac{\text{Strength of I-section}}{\text{Strength of circular section}} = \frac{Z_1}{Z_3} = \frac{1543800}{178502.9} = 8.649$$

Exercise

5.1 Derive the expression for bending stress for a layer in a beam.

5.2 What do you mean by section modulus? Find an expression for section modulus for rectangular, circular and hollow circular section.

5.3 What is the meaning of 'strength of a section'?

5.4 What do you mean by 'simple bending' or 'pure bending'? What are the assumptions made in theory of simple bending?

5.5 A timber beam is 120 mm wide and 200 mm deep and is used on a span of 4m. the beam carries a udl of 2.8 kN/m run over the entire length. Find the maximum bending stress induced.

(Ans. 7 N/mm²)

5.6 A rectangular beam 10 m long is simply supported at its ends and is acted upon by a 10 kN/m udl throughout its length. Find the size of the beam if the thickness is twice that of its width. Take the allowable bending stress as 40 N/mm².

(Ans. $b = 167.3$ mm, $d = 334.6$ mm)

5.7 A rectangular beam 300mm deep is simply supported over a span of 4m. what udl per meter the beam may carry if the bending stress is not to exceed 120 N/mm². Take $I = 8 \times 10^6$ mm⁴.

(Ans. 3.2 kN/m)

5.8 For a given stress compare the moments of resistance of a beam of square section when placed (a) with its two sides horizontal (ii) with its diagonal horizontal.

(Ans. 1.414)

5.9 A 10m long C.I. pipe of 500mm internal diameter and 20 mm thickness is simply supported at its ends. Find the maximum intensity of bending stress induced in the metal when the pipe is running full with water. Take unit weights of C.I. and water as 70 kN/mm³ and 10 N/mm³.

(Ans. 13.17 N/mm²)

5.10 A beam of I-section 400mm deep has flanges 200 mm wide and 30 mm thick and web 24 mm thick. Compare the flexural strength with that of a beam rectangular section of the same weight, depth being twice the breadth.

(Ans. 3.27)